

Use the following context to answer questions 1-5.

The length of a machine part should be 5.25 inches (in). The manufacturing tolerance is within 0.03 in.

1. What is the minimum value the machine part could be?
2. What is the maximum value the machine part could be?
3. Sketch a number line that shows the minimum value, maximum value, and 5.25.



4. Write an absolute value equation that could be used to determine the maximum and minimum value of m , the length of the machine part, algebraically.
5. Show that the solutions to the equation match the maximum and minimum values determined in questions 1 and 2.

For questions 6-10, solve each absolute value equation.

6. $0 = 3|x + 2| - 15$

7. $|-4x + 2| - 9 = 0$

8. $|x + 5| + 9 = 3$

9. $|3x + 4| - 6 = 25$

10. $-|x - 3| + 7 = 0$